

Name

Activity 3.6 - Practical Skills in Biology.

In this task you will carry out some of the activities of scientists to collect and record information. Complete all your written answers in the spaces provided on the paper.

1. Using a microscope: Go to Station 1 **(4 marks)**
 - a) Focus on the slide using the low power (X10) objective. **Put your hand up for your teacher to mark this step.**
 - b) While your teacher is watching, focus under high power.
 - c) Calculate the total magnification under high power and write your answer below:

2. Making a wet mount: Go to Station 2 **(5 marks)**
 - a) Using the equipment provided, make a stained wet mount of a small piece of the specimen provided. You may use the microscope at this station to check that you are happy with your prepared slide.
 - b) When you have finished, write your name on a label and stick it on one end of your slide. Your teacher will mark this slide later.

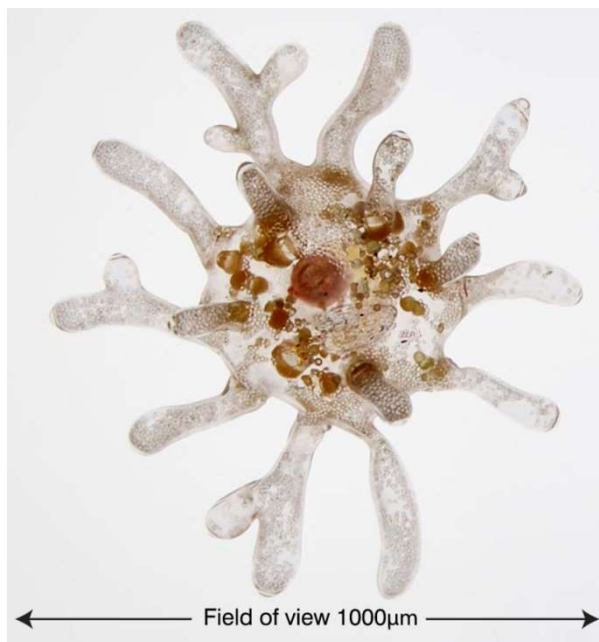
3. Table of results **(4 marks)**

Ayşe wanted to compete in the high jumping at the next school athletics carnival. She hypothesized that if she practiced her jumping every day for a week she should improve her maximum height jumped. On Monday she managed to clear 1.2 m. On Tuesday she cleared 1.4 m, Wednesday 1.2 m, Thursday 1.4 m, Friday 1.6 m, Saturday 1.5 m and Sunday 1.7 m.

Make a table of results for Ayşe with an appropriate heading.

Practical Skills in Biology (continued)

- c) Here is a picture taken through a microscope of a microorganism. Is this a plant or animal? Explain. **(2 marks)**



- d) If the diameter of the field of view of the microscope is 1000µm, calculate the width of the body part of the organism (ignore the appendages). **(2 marks)**

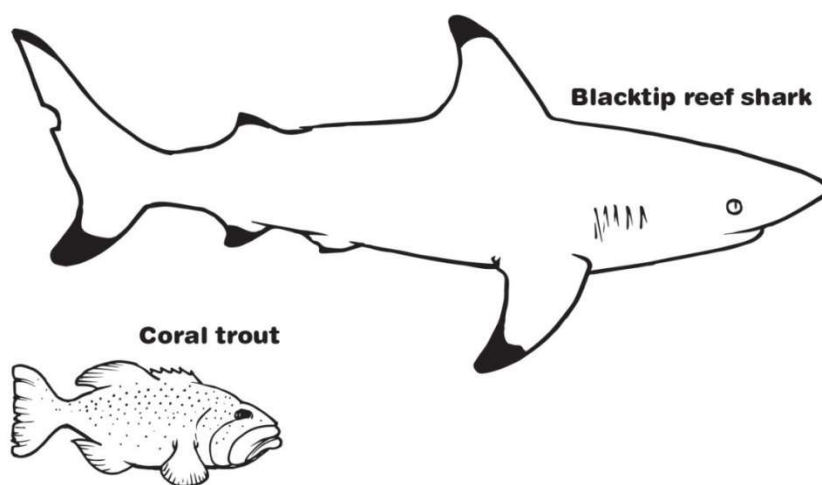
Practical Skills in Biology (continued)

5. Scale drawings

(6 marks)

- a) The diagram below shows two animals found in the Great Barrier Reef; the coral trout and the blacktip reef shark. The two animals are drawn to scale. If the blacktip shark is 1.5 metres long, use these diagrams to calculate how long the coral trout is. Show your working.

(2 marks)



- b) Your teacher will give you a part of a living organism. In the space below, make a biological drawing of this organism, two times the actual size. **(4 marks)**

Activity 5.9 - Topic Test

Part A: Multiple-choice questions

(1 mark each)

On the answer sheet mark the letter corresponding to the correct answer with an X.

1. Living things react to stimuli. What are stimuli?
 - A. types of food
 - B. movement of limbs
 - C. predators
 - D. changes in the environment.

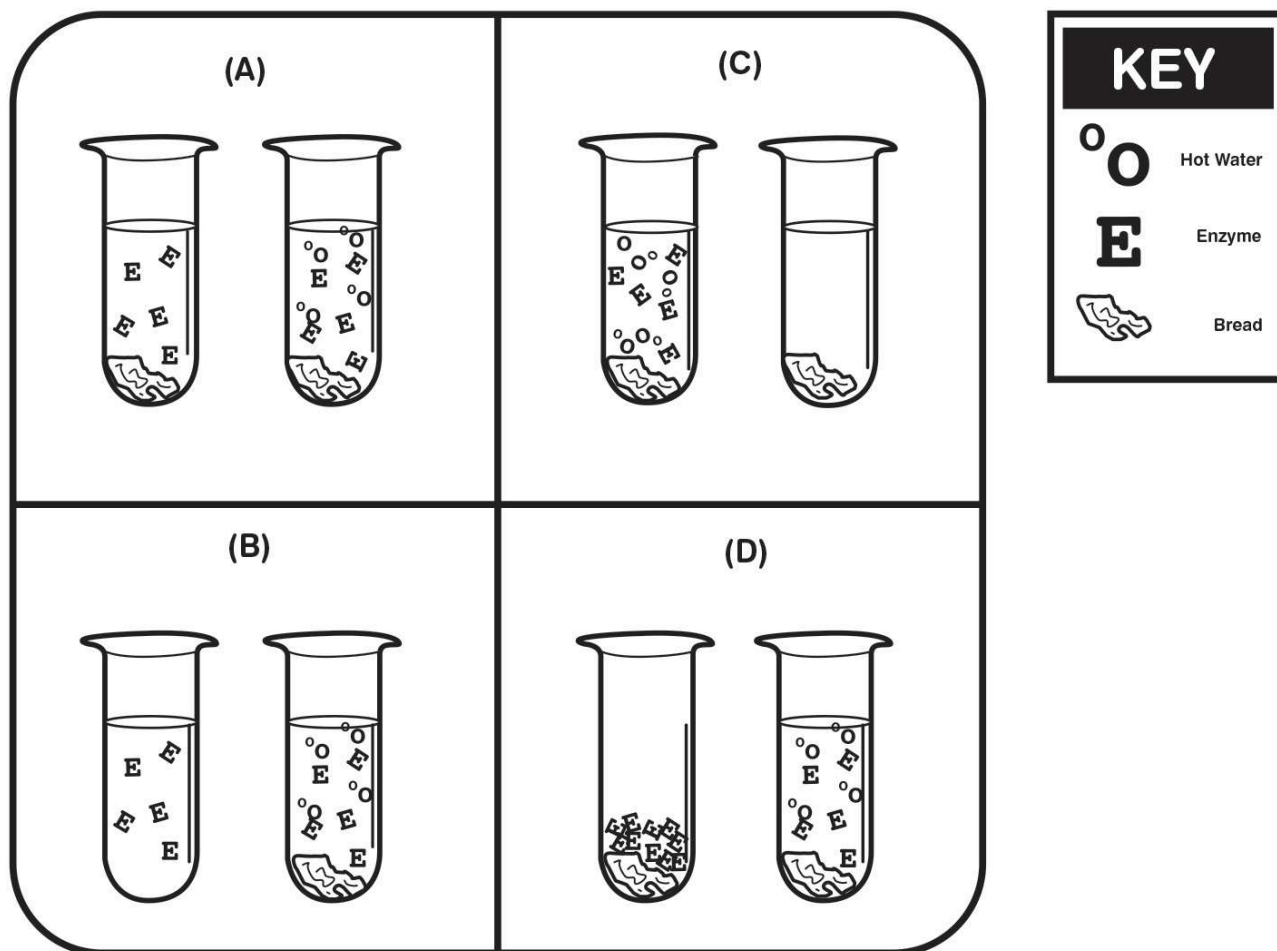
2. Which of the following is found in all animal cells?
 - A. vacuole
 - B. cell membrane
 - C. cell wall
 - D. chloroplasts.

3. Which of the following statements about cells is correct?
 - A. chloroplasts are found in all plant cells
 - B. not all living things contain cells
 - C. the nucleus contains inherited material
 - D. cytoplasm is only found in plant cells.

4. The cells of bacteria differ from plant and animal cells because they:
 - A. cannot live independently
 - B. do not contain a nucleus
 - C. contain mitochondria
 - D. do not have a cell wall.

Topic Test (continued)

5. During digestion food is broken down with the help of chemicals called enzymes. If you wanted to find out the effect of high temperature on these enzymes, which two test tubes would you need to set up?



6. Programmed cell death and renewal is an advantage to living organisms because:
- death is a normal process
 - cells all die after a few months
 - cells cannot survive for the whole life of the individual
 - cell division occurs in all tissues.
7. Sexual reproduction differs from asexual reproduction because:
- it involves the fusion of gametes
 - the offspring are all identical
 - it always involves mating
 - it occurs much more quickly.

Topic Test (continued)

8. Which of the following is an advantage of external fertilization in fish?

- A. a smaller number of eggs are required
- B. a smaller number of sperm are required
- C. the chances of survival are greater
- D. no parental care is needed.

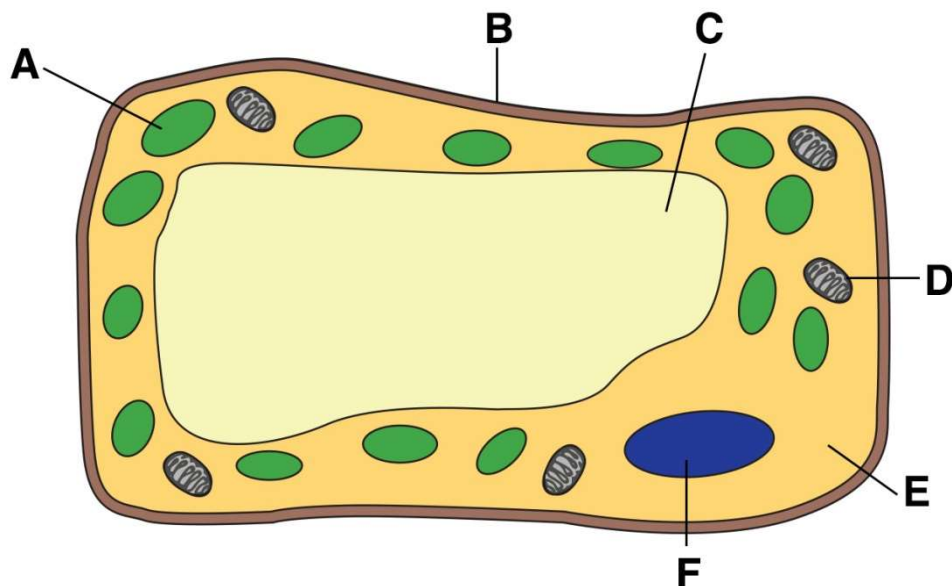
9. A major advantage of sexual reproduction is:

- A. variation in offspring
- B. a larger number of offspring
- C. healthier offspring
- D. faster development of offspring.

10. Asexual reproduction is an advantage when:

- A. growth conditions are poor
- B. a variety of offspring is needed
- C. identical offspring are required
- D. a disease is likely to attack the population.

11. (6 marks) Look at the following cell diagram.



In the table on your answer sheet, name the parts A to F and describe a function of each.

Topic Test (continued)

12. (2 marks)

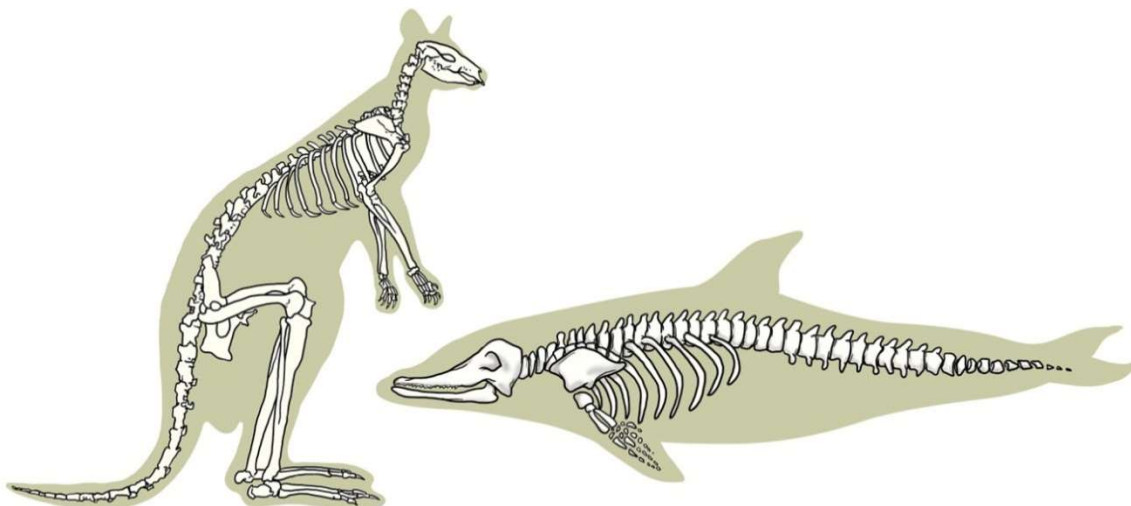
- What do we mean when we say that cells specialise?
- Give an example to explain your answer

13. (4 marks)

Explain the relationship between, tissues, organs and systems. Give examples.

14. (4 marks)

Dolphins and kangaroos are both mammals with similar patterns in their skeletal structure. Yet there are major differences.



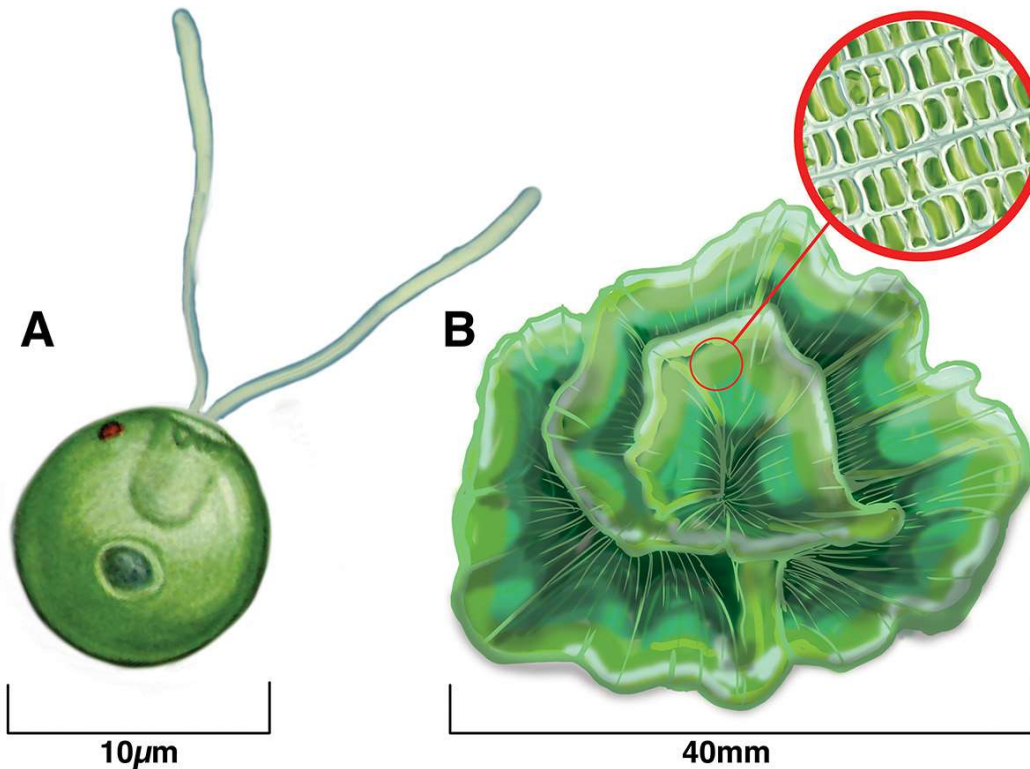
Complete the table on your answer sheet to relate the differences in structure to the lifestyle of each animal.

Topic Test (continued)

15.

(4 marks)

Consider the two species of algae, A and B.



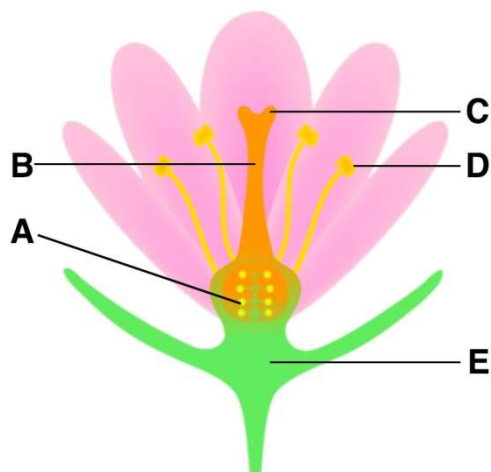
- Calculate how much bigger alga B is compared to alga A.
- Why must single-celled organisms remain small to survive?
- What happens when a single-celled organism grows to its maximum size?
- What advantage is there in being multicellular?

Topic Test (continued)

16.

(5 marks)

Look at the diagram of the flower to answer questions.



- Male gametes form at:
A B C D E
- Pollination is the transfer of pollen to:
A B C D E
- Fertilisation occurs at:
A B C D E
- Describe one adaptation of a flower and how it assists the transfer of pollen by a particular pollinator.

17.

(5 marks)

The discoveries of many scientists have had an impact on human activities, such as medicine, agriculture and industry. In this topic, you have studied the work of a few of these people.

Choose **one** of the following scientists: Robert Brown, Antonie van Leeuwenhoek, Robert Hooke or Fiona Wood

For the scientist of your choice, answer the following questions;

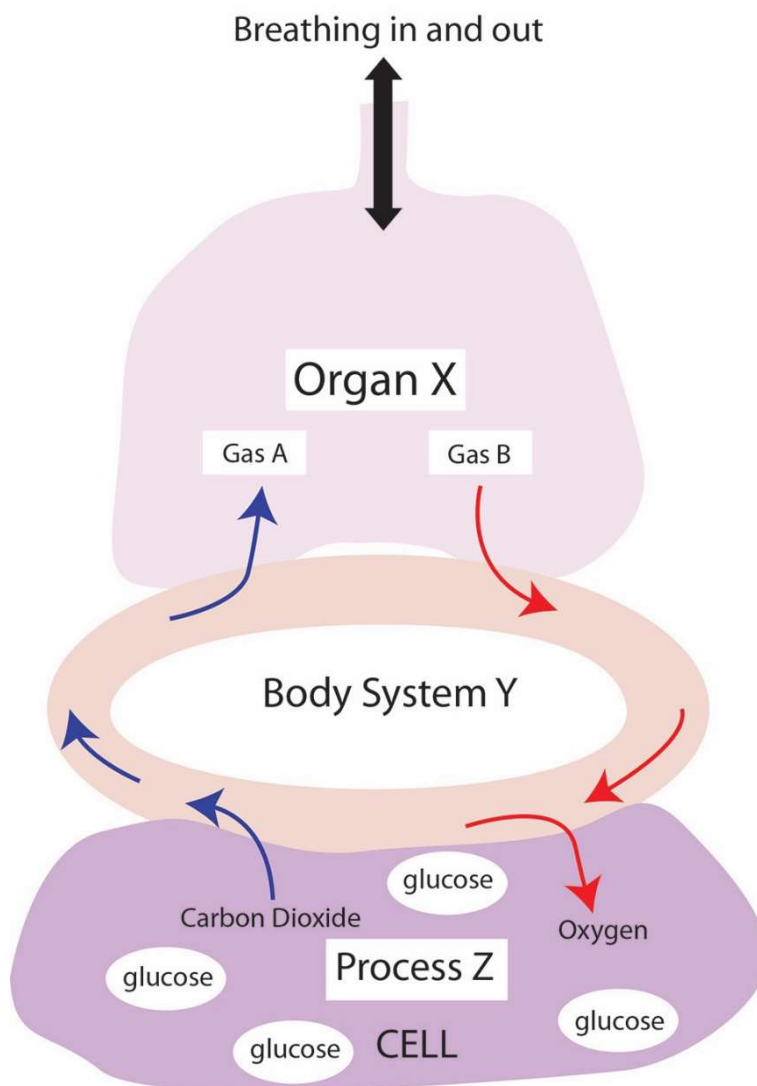
- Describe the area of science that the scientist worked in.
- What did this scientist discover?
- What was the effect of this work on scientific progress at the time?
- What impact did/will this scientist's work have in the future?

Topic Test (continued)

18.

(6 marks)

Look at the following diagram representing processes occurring in the human body.



Name the following:

- organ X
- body system Y
- process Z
- gas A
- gas B
- What is the purpose of process Z?

Topic Test (continued)

19.

(10 marks)

Read the article below to answer this question:

Is there life on the planet Novaterra?

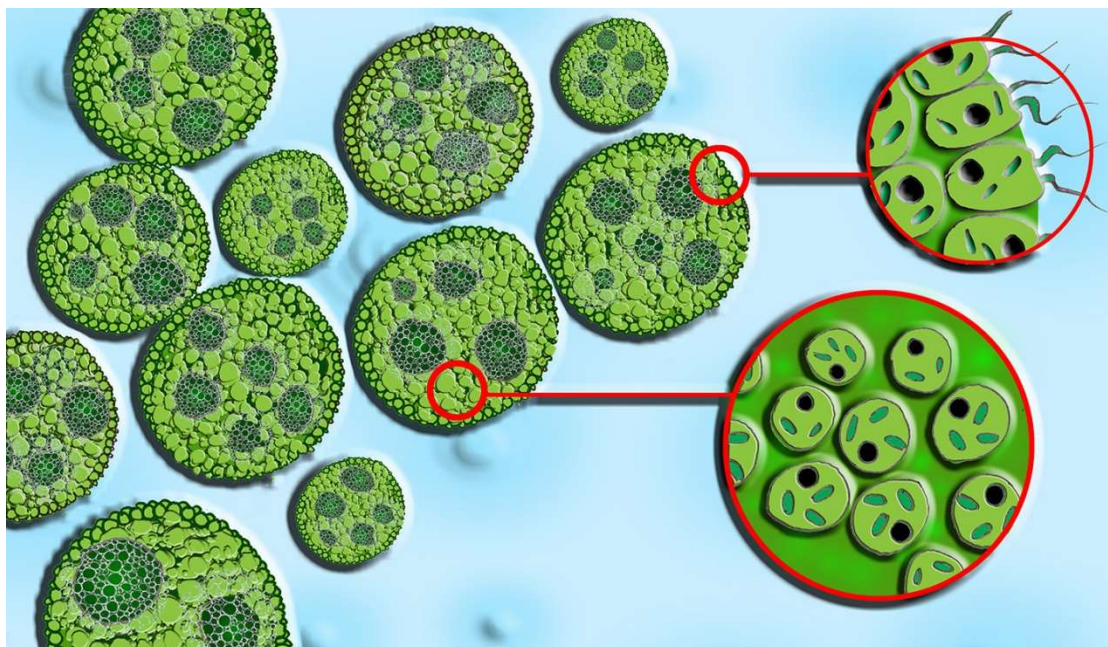
The discovery of a new planet, Novaterra, which may have a similar environment to Earth has given scientists hope of finding clues to life in outer space.

Earth-like geological formations and the presence of pools of water caused great excitement for the scientists. They then took water samples and observed them under a microscope.

They discovered spherical objects, which were green in colour and appeared to move randomly in the water. They named these 'globs'. On closer observations, many of the globs were seen to have several smaller spherical structures inside them.

The scientists kept several globs for a week in a tank to investigate them in more detail. They were surprised to find the number of globs had increased dramatically in this time, so they decided to look at one glob in more detail. With continuous observation they were able to see small spherical structures pushing their way out of the glob and growing to become independent globs. They also observed that globs moved towards a light shone into the tank.

Greater magnification showed the globs were made up of smaller units, as shown in the picture below. Some units on the outside of the globs had hair-like protrusions.



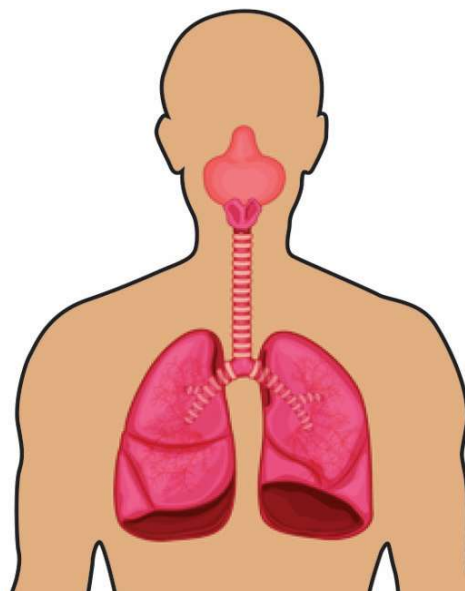
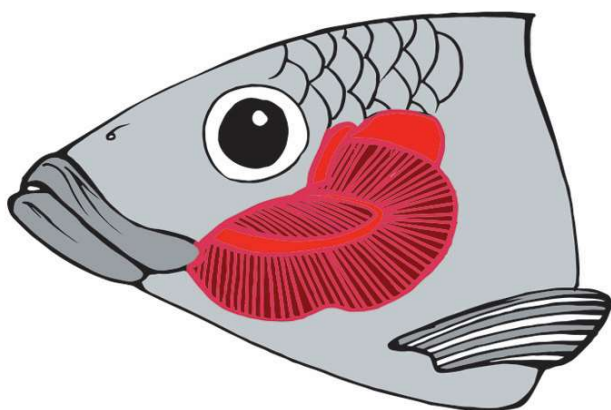
Topic Test (continued)

- Why did the presence of water make the scientists excited?
- State two characteristics of life that the globs show.
- If globs are living, how do you think they might gain energy for life processes?
- Can you be certain that globs are living things? Discuss this question using information from the article.
- Describe two further tests the scientists could do to confirm whether globs are living?

20.

(4 marks)

The diagrams below illustrate the respiratory systems of a fish and a mammal.



- Give one similarity between these two systems.
- What is the major difference in the arrangement of the organs in these systems?
- Why would a mammal such as a human not survive under water?

Topic Test (continued)

21.

(14 marks)

Professor Barry Marshall is an Australian scientist who was awarded the Nobel Prize for Medicine for his discovery of the cause and treatment for gastritis and stomach ulcers. He tested many patients with ulcers and found that the bacterium, *Helicobacter pylori*, was present in all cases. He also found that these bacteria could be killed using antibiotics.

To prove that the bacteria was causing the ulcers, he performed an experiment on himself. After checking that he did not already have a stomach ulcer or gastritis, he drank a solution containing *Helicobacter* bacteria. After several days he developed the symptoms of gastritis and a test done after 10 days confirmed that his stomach had been infected with the bacteria.

- Before carrying out his experiment, Barry Marshall had an endoscopy to show that he did not have stomach ulcers.
What is an endoscope and why was this invention an important advance in medicine?
- Explain two ways in which Barry Marshall's experiment was well designed.
- Did this experiment conclusively prove that ulcers were caused by the *Helicobacter pylori* bacteria? Give two reasons in support of your answer.
- A scientist wanted to find out how quickly bacteria would grow in the stomach. She designed an experiment to test this by growing the bacteria in a solution similar to that found in the stomach, at 37°C. Her results are set out in the table below:

Time (hours)	Number of bacteria (1000s per cm ³)
2	10
4	13
6	18
8	24
10	32
12	44

- On the graph paper provided, draw a graph of these results.
- Describe the trend in number of bacteria over this period.
- If the bacteria are multiplying by asexual reproduction, what does this tell you about the new cells produced?
- Do you think that the bacteria will continue to grow and reproduce in this way? Explain.

Activity 5.9: Topic Test - Answer sheet

Total mark
/75

Part A: Multiple-choice questions.

Mark **X** in the box corresponding to the correct answer.

Question	A	B	C	D
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				

Topic Test - Answer sheet (continued)

11.

(6 marks)

	Name of part	Function
A		
B		
C		
D		
E		
F		

12.

(2 marks)

- a)
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- a)
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Topic Test - Answer sheet (continued)

13. (4 marks)

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14. (4 marks)

Part of skeleton	Kangaroo	Dolphin
Forelimb		
Hind limb		
Tail		
Backbone		

Topic Test - Answer sheet (continued)

15. (4 marks)

- a)
-
- b)
-
- c)
-
- d)
-

16. (5 marks)

Question	A	B	C	D
a)				
b)				
c)				

- d)
-

Topic Test - Answer sheet (continued)

17.

(6 marks)

- a)
- b)
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- c)
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- d)
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18.

(6 marks)

- a)
- b)
- c)
- d)
- e)
- f)

Topic Test - Answer sheet (continued)

19 a) (1 mark)

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b) (2 marks)

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c) (1 mark)

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d) (4 marks)

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e) (2 marks)

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20. (4 marks)

a)

b)

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c)

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Topic Test - Answer sheet (continued)

21.

a) (2 marks)

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b) (2 marks)

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c) (2 marks)

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d) (i) Use graph paper provided. (5 marks)

(ii) (1 mark)

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(iii) (1 mark)

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(iv) (1 mark)

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